Design and Evaluation of a Machine Learning Workflow for Breast Cancer Diagnosis Classification

Elif Güler, November 2026 (revised version)

Abstract

Accurate classification of breast masses as malignant or benign supports timely treatment and can reduce unnecessary biopsies. This reproducible project uses the Wisconsin Diagnostic Breast Cancer (WDBC) dataset to study exploratory analysis, feature redundancy, feature selection and the comparison of seven classifiers. In this revised version, scaling, feature selection and the classifier are always fitted together inside a scikit-learn pipeline within each cross-validation fold, so that validation labels cannot influence preprocessing. Without leakage, selecting a subset with SelectKBest, RFE or their intersection does not improve on the full feature set and these subsets still contain 16–21 strongly correlated feature pairs. A correlation filter with threshold 0.90 removes the redundancy (21 features) at almost no cost (97.3 % CV accuracy). On the held-out test set (114 samples), tuned Logistic Regression reached 98.2 % accuracy (2 missed malignant cases, no false positives, MCC 0.96, AUC 0.998), followed by the RBF-kernel SVM and Random Forest (97.4 %). The differences between the best models correspond to one or two test samples and are not statistically meaningful.

1. Introduction

Breast cancer is one of the most commonly diagnosed cancers, and early and accurate diagnosis strongly affects treatment outcomes. Visual interpretation of fine needle aspirate (FNA) biopsies is time consuming and can differ between observers (Adedayo-Ajayi et al., 2023). With digitised cytological measurements, statistical classification is increasingly used to support the traditional procedure (Casaubon et al., 2026). Machine learning is well suited to this problem because the boundary between malignant and benign cells is described by several geometric measurements rather than a single threshold (Sharma et al., 2018). The WDBC dataset (Wolberg et al., 1993) is a standard benchmark for this binary classification problem.

The aims of this project are (i) to identify which classifier performs best on this data and why, (ii) to examine the redundancy in the 30 features, and (iii) to test whether a smaller feature subset can match the performance of the full set. The revised version focuses on a methodologically correct validation: every data-dependent step is part of the cross-validated pipeline.

2. Materials and Methods

2.1 Dataset

WDBC contains 569 images of FNA samples of breast masses (357 benign, 212 malignant, i.e. 62.7 % and 37.3 %). Ten nucleus properties (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension) are each summarised as mean, standard error (se) and worst value, giving 30 features. There are no missing values, so no imputation was needed. Malignant is the positive class. Because the classes are moderately imbalanced, F1, MCC, AUC and the numbers of false negatives and false positives are reported in addition to accuracy.

2.2 Split and pipelines

The data were split once, stratified, into 80 % training (455 samples) and 20 % test (114 samples, 42 malignant) with `random_state=42`, before any exploration or fitting. Exploratory analysis uses the training set only. The features have very different scales (the standard deviation of `area_mean` is

about 360, that of smoothness_mean about 0.014), so all models are scikit-learn pipelines of the form StandardScaler → (optional feature selector) → classifier. Scaler and selector are fitted only on the training part of each fold, and the test set is used once at the end.

2.3 Feature selection experiments

All feature selection experiments use Logistic Regression and repeated stratified 5-fold cross-validation (3 repeats, 15 scores per estimate). The compared strategies are: SelectKBest (ANOVA F-statistic) and Recursive Feature Elimination (RFE with logistic regression) for k = 5…30, with the smallest k within one standard error of the best chosen (1-SE rule) and nested CV as an unbiased estimate; the intersection of both (the subset of the first version, k = 19 and k = 17); and a correlation filter, which visits features from the most to the least discriminative and keeps a feature only if its absolute correlation with all kept features is at most 0.90 (or 0.95). The redundancy of a subset is measured as the number of feature pairs with $|r| > 0.9$.

2.4 Classifiers, tuning and evaluation

Seven classifiers were compared: Logistic Regression (LaValley, 2008), Linear Discriminant Analysis (Xanthopoulos et al., 2012), k-Nearest Neighbors (Kramer, 2013), Support Vector Machine (Suthaharan, 2016), Gaussian Naive Bayes (Reddy et al., 2022), CART (Ying, 2015) and Random Forest (Cutler et al., 2012), implemented in scikit-learn (Pedregosa et al., 2011). Six were tuned with GridSearchCV (5-fold, accuracy); Gaussian Naive Bayes has no hyperparameters to tune and was fitted with default settings. The grids contain 18 (Logistic Regression), 2 (LDA), 20 (k-NN), 18 (SVM), 24 (CART) and 72 (Random Forest) settings; thus the SVM grid is smaller than the Random Forest grid. For the SVM, GridSearchCV was also compared with RandomizedSearchCV (8 of 18 settings). Tuned models were evaluated once on the test set with accuracy, precision, recall (sensitivity to malignant cases), F1, MCC, AUC, and the numbers of false negatives (FN, missed malignant cases) and false positives (FP).

Three kinds of results are reported and never mixed: (a) cross-validation results on the training set (feature-selection study and default models), (b) tuned-model results on the test set with all 30 features, and (c) tuned-model results on the test set with the reduced feature set.

3. Results

3.1 Exploratory analysis

Skewness. 23 of the 30 features have a skewness above 1, and all ten se features are strongly right-skewed; the most extreme are area_se (5.5), concavity_se (5.4) and fractal_dimension_se (4.1). Standardisation does not remove the resulting high outliers, so scale-sensitive models can be affected by them; tree-based models are unaffected by skew.

Class differences. The most discriminative features, measured by Cohen's d between malignant and benign, are concave_points_worst (2.62), perimeter_worst (2.60), radius_worst (2.55), concave_points_mean (2.54), perimeter_mean (2.27) and area_worst (2.21). Malignant nuclei are larger and have more concave contour points, and the boxes of the two classes barely overlap for these features (Figure 1). At the other end, texture_se, symmetry_se, fractal_dimension_mean and smoothness_se have $|d|$ close to 0 and carry little information on their own. Outlier counts are small and show no consistent pattern: area_worst has more outliers in malignant cases (7 against 3), but concave_points_mean has more in benign cases (12 against 6).

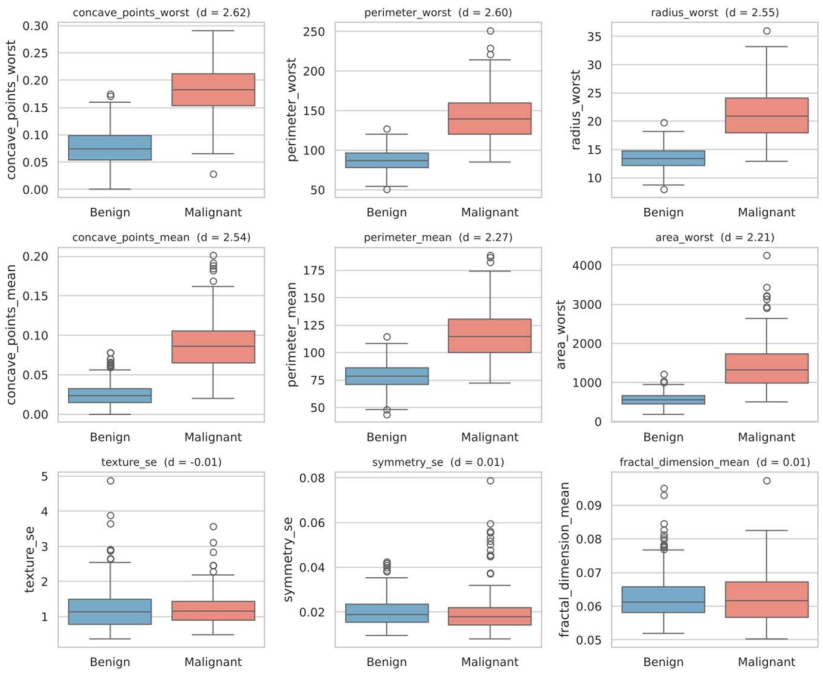


Figure 1. Boxplots by diagnosis of the six most and the three least discriminative features (d = Cohen's d, training set).

Redundancy. There are 21 feature pairs with |r| > 0.90 and 15 with |r| > 0.95. The strongest are geometric by construction (radius–perimeter 0.998 for the mean features and 0.994 for the worst features; radius–area 0.987), and between the mean and worst versions of a property (perimeter 0.97). Figure 2 shows the block of strongly correlated size and concavity features. Such redundancy inflates the variance of coefficients in linear models and makes them hard to interpret, but it hurts prediction accuracy much less.

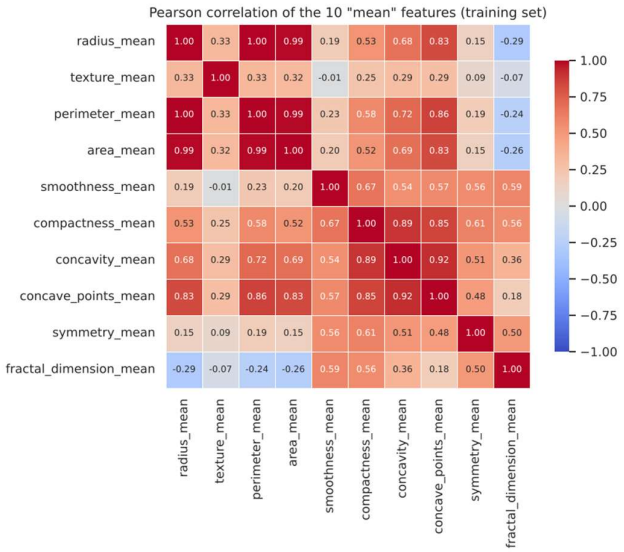


Figure 2. Pearson correlation of the ten "mean" features (training set).

PCA (used for visualisation only, fitted on the scaled training data without labels): the first component explains 44.6 % of the variance and the first two 63.1 %; 10 components are needed for 95 %, which again reflects the redundancy. In two dimensions (Figure 3) the classes are largely separated, with a transition zone where they overlap. The problem is therefore close to linearly separable, but not perfectly so.

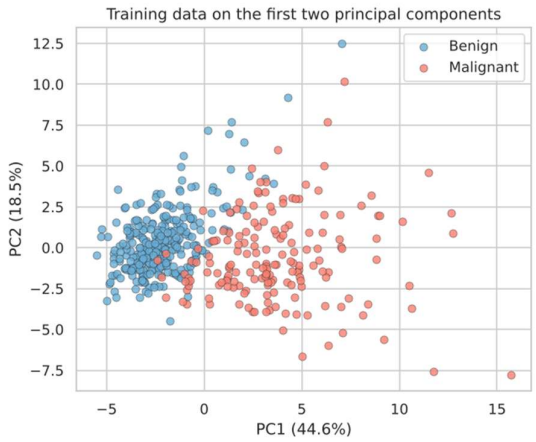


Figure 3. Training data on the first two principal components.

3.2 Feature selection

Scaling and selection before cross-validation, as in the first version, make the CV estimate optimistic (Table 1): the leaky estimate is higher by up to 0.7 percentage points at k = 16, the size of the subset used in the first version.

Table 1. CV accuracy of SelectKBest + Logistic Regression: selection before CV (leaky) vs. inside the pipeline.

k	Leaky (%)	Pipeline (%)	Optimism (pp)
5	94.2	94.2	0.00
10	94.8	94.7	0.07
16	96.3	95.7	0.66
20	97.2	96.9	0.29
25	97.5	97.5	0.00

Inside the pipeline (Figure 4), SelectKBest needs about 24 features to reach the accuracy of the full set and is clearly worse for k ≤ 15 (94–95 %), whereas RFE reaches about 96.5 % with only 5–10 features and peaks at 97.8 % for k = 24 (1-SE rule: k = 18). SelectKBest ranks features one by one, so the near-identical scores of radius, perimeter and area take most of the first slots without adding information; RFE handles redundancy better. With k chosen inside the loop, nested CV gives 97.5 % (SelectKBest) and 97.3 % (RFE) with standard deviations of 1.6–1.9 pp, which is no gain over all 30 features (97.6 %).



Figure 4. CV accuracy of Logistic Regression as a function of the number of selected features k (selection inside the pipeline).

Table 2. Feature sets (Logistic Regression, repeated 5×3 CV on the training set).

Feature set	Features	CV acc. (%)	Std (pp)	Pairs |r|>0.9	Max |r|
All 30 features	30	97.6	1.4	21	0.998
SelectKBest ∩ RFE (original subset)	16	96.8	1.6	16	0.994
SelectKBest, k = 24 (1-SE rule)	24	97.6	1.5	21	0.998
RFE, k = 18 (1-SE rule)	18	97.7	1.2	16	0.994
Correlation filter, |r| ≤ 0.95	23	97.2	1.4	3	0.921
Correlation filter, |r| ≤ 0.90	21	97.3	1.2	0	0.897

The SelectKBest ∩ RFE subset of the first version reaches 96.8 % (± 1.6 pp) against 97.6 % for all 30 features. The loss of 0.8 pp is within the fold-to-fold spread, so the subset is not clearly worse, but it is not better either. It is dominated by the size family (radius, area and perimeter together), plus texture, concavity and concave points, and contains no compactness or fractal-dimension feature. It therefore keeps exactly the group with the highest redundancy, and 16 strongly correlated pairs remain; agreement of two selectors does not mean that the subset is optimal or free of redundancy.

The correlation filter keeps only radius_se and perimeter_worst from the size group, drops area completely and keeps nearly all shape descriptors (smoothness, compactness, concavity, symmetry, fractal dimension). It is the only strategy that removes the redundancy (0 pairs above 0.9) and it costs only 0.3 pp of CV accuracy, well inside one standard deviation. It is used as the reduced feature set below.

3.3 Default models

Table 3. Default hyperparameters, all 30 features (repeated 5×3 CV on the training set).

Model	CV accuracy (%)	Std (pp)
Logistic Regression	97.6	1.4
Support Vector Machine (RBF)	97.1	1.5
K-Nearest Neighbors	96.4	2.4
Random Forest	95.8	2.7
Linear Discriminant Analysis	95.3	2.0
Gaussian Naive Bayes	93.9	2.6
CART	92.2	3.0

With default settings the top models are statistically indistinguishable (Figure 5); only the gap to CART and Gaussian Naive Bayes is clear. Note that the SVM is a nonlinear model (RBF kernel by default); the linear models in this comparison are Logistic Regression and LDA.

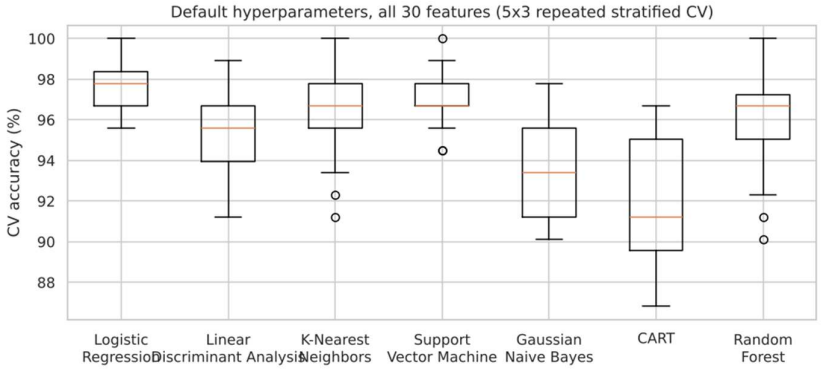


Figure 5. Default hyperparameters, all 30 features: 15 CV scores per model (5-fold, 3 repeats).

3.4 Tuned models on the test set

Both searches for the SVM (GridSearchCV and RandomizedSearchCV) reached 97.4 % CV accuracy and selected an RBF kernel; the randomised search needs fewer fits (8 instead of 18 settings) and is somewhat faster, but on a grid this small the difference is negligible. The selected hyperparameters with all 30 features were: Logistic Regression C = 0.1, L2 penalty, liblinear solver; SVM RBF kernel, C = 10, gamma = scale; Random Forest 100 trees, maximum depth 5, sqrt features; k-NN k = 3 with Manhattan distance; CART entropy criterion, maximum depth 5, minimum split 10; LDA with the SVD solver.

Table 4. Tuned models, all 30 features, held-out test set (114 samples, 42 malignant).

Model	CV acc. (%)	Test acc. (%)	Prec. (%)	Recall (%)	F1 (%)	MCC	AUC	FN	FP
Logistic Regression	97.4	98.2	100.0	95.2	97.6	0.96	0.998	2	0
Support Vector Machine	97.6	97.4	100.0	92.9	96.3	0.94	0.993	3	0
Random Forest	96.7	97.4	100.0	92.9	96.3	0.94	0.995	3	0
Linear Discriminant Analysis	95.6	96.5	100.0	90.5	95.0	0.93	0.997	4	0
K-Nearest Neighbors	97.6	96.5	100.0	90.5	95.0	0.93	0.974	4	0
CART	93.0	95.6	97.4	90.5	93.8	0.91	0.966	4	1
Gaussian Naive Bayes	93.8	92.1	92.3	85.7	88.9	0.83	0.989	6	3

Table 5. Tuned models, reduced feature set (21 features, correlation filter 0.90), held-out test set.

Model	CV acc. (%)	Test acc. (%)	Recall (%)	MCC	AUC	FN	FP
Support Vector Machine	96.9	97.4	92.9	0.94	0.993	3	0
Linear Discriminant Analysis	95.2	96.5	90.5	0.93	0.996	4	0
Random Forest	96.7	96.5	90.5	0.93	0.995	4	0
Logistic Regression	97.4	96.5	92.9	0.92	0.997	3	1
K-Nearest Neighbors	96.7	94.7	88.1	0.89	0.953	5	1
CART	91.9	93.0	88.1	0.85	0.920	5	3
Gaussian Naive Bayes	91.9	89.5	83.3	0.77	0.976	7	5

With the reduced set the SVM and LDA are unchanged, Logistic Regression and Random Forest lose one to two test samples, and k-NN, CART and Gaussian Naive Bayes lose more. One test sample corresponds to 0.9 pp of accuracy and 2.4 pp of recall, so most of these differences are within noise.

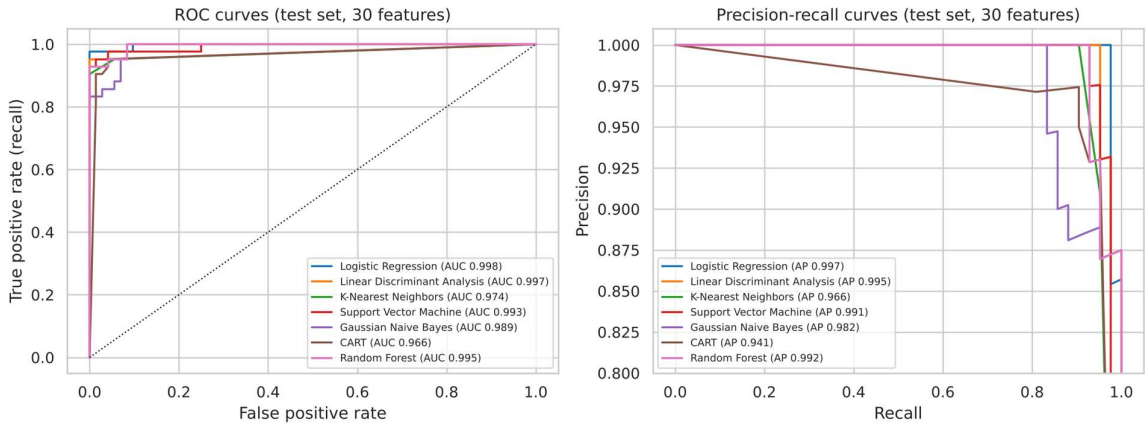


Figure 6. ROC and precision–recall curves of the tuned models on the test set (30 features).

Errors. Apart from Gaussian Naive Bayes and CART, no model produced a false positive; all errors are malignant cases predicted as benign (Figure 7): 2 of 42 for Logistic Regression, 3 for the SVM and Random Forest, 4 for LDA and k-NN. In a clinical setting this is the costly error, because a missed malignant case may delay treatment. Sensitivity should therefore be reported together with accuracy, and the decision threshold could be lowered if sensitivity matters most (not tuned here).

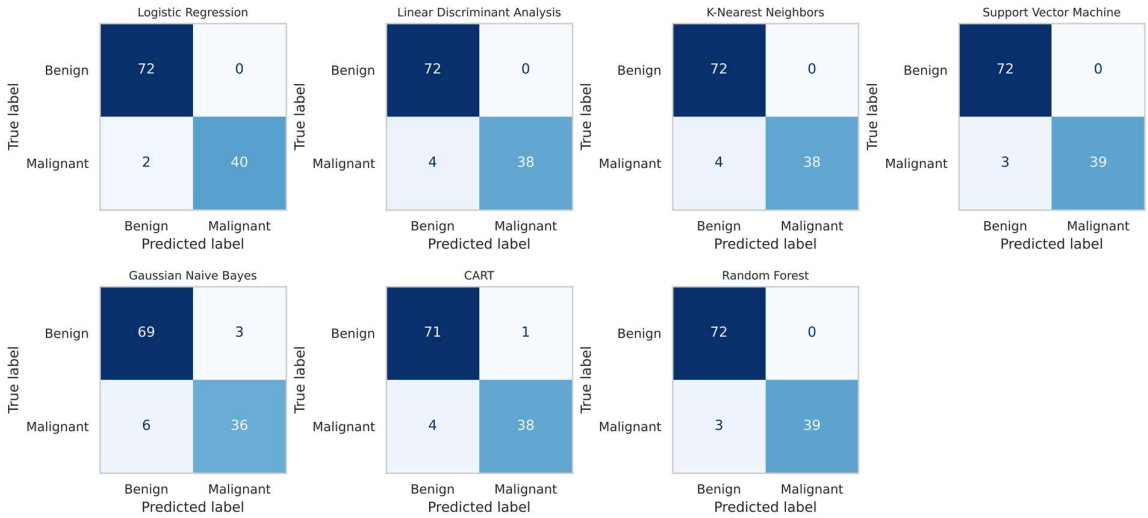


Figure 7. Confusion matrices of the tuned models on the test set (30 features).

4. Discussion

Once the leakage is removed, feature selection does not improve prediction on this data: the best selected subsets are as good as, but not better than, the full set, and SelectKBest, RFE and their intersection keep many redundant features because they favour the same large-scale geometric measurements. Redundancy is better handled explicitly, with a correlation filter, which gives a non-redundant 21-feature set with practically unchanged CV accuracy. Its test accuracy is slightly lower for several models, so the full feature set remains a safe default and the reduced set a reasonable choice if interpretability matters.

Logistic Regression and the nonlinear RBF SVM perform about equally well, while the other linear model (LDA) is slightly behind and the more flexible Random Forest does not beat the simpler models. The data are close to linearly separable (PCA), but the results do not show that nonlinear models are worse: the selected SVM is nonlinear and performs at the same level.

5. Conclusion

A correctly validated pipeline identifies Logistic Regression (98.2 % test accuracy, 95.2 % recall, 100 % precision, AUC 0.998) as the best model, with the RBF SVM and Random Forest very close (97.4 %); Gaussian Naive Bayes and CART are clearly weaker. The advantage of Logistic Regression is only one or two test samples and should not be over-interpreted. Feature selection with SelectKBest or RFE gave no benefit over the full feature set, and only a correlation filter removed the redundancy.

Limitations. The dataset is small (114 test samples) and comes from a single source, and the metrics rest on a single 80/20 split; repeated or nested CV for all tuned models would give tighter comparisons. No transformation of the skewed features and no threshold tuning were tried. An independent dataset would be needed to confirm that the results generalise.

Data availability

All code, figures, tables, trained pipelines and the notebook are available at https://github.com/elif-guler/breast-cancer-ml. Please check the README page and follow the usage guide to replicate the results.

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